# Sources of capital growth

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ABSTRACT

Data from national accounts support an argument that capital growth and acceleration arrive without help from net saving. We explore ways in which this is possible, and discuss implications for economic teaching and public policy.

## 1. Introduction and overview

Many economists over the centuries have reasoned that net saving[1] should tend to give equal capital growth. Economists since the early nineteenth century have added the proviso that net saving cannot safely outpace innovation; more capital must mean capital redesigned for greater productivity if economies are to escape risk of capital glut and diminishing returns (West (1815), Ricardo (1815), Malthus (1815)). Roy Harrod (1939)) described that limit for safe net saving, meaning the rate of imagining and developing new ideas for more productive forms of capital, as the "warranted rate". Harrod, and many other economists of his time and since, have focused on growth of output rather than of capital, but have modeled growth of output by first assuming the equivalence of net saving and capital growth, within the warranted rate, and then looking for effects of that capital growth on later output growth.

Some other economists, including John Rae (1834) and John Stuart Mill (1848) (see Appendix C), argued that capital growth might also be explained by a rise in productivity of capital and labor already extant. Ways might be found for existing factors to produce more, that is, and so to allow more consumption, or more capital growth, or any mix of the two, without inputs of net saving. Robert Solow (1957) allowed that possibility for "disembodied" growth, where plant and products already existing are repurposed or redeployed in more productive ways.

We test between those two explanations of capital growth, by net saving or by increase in productivity of capital and labor already in existence, by comparing net saving to concurrent change in market-value capital in 92 countries, and also comparing changes in those two variables. All data are drawn from national accounts of those countries as collated on the free website World Inequality Database. Our findings are easiest to reconcile in an interpretation that net saving adds nothing to capital growth.[2] Net saving, if so, raises the physical quantity of capital, but not the aggregate value, and so reduces the value per unit. This condition describes capital glut.

The data are most easily explained by the present value principle, and by production efficiencies enabled through innovation. Value is created in the mind of the market at the moment when prospective cash flows are discounted. It is created only if the market sees a path, step by step, from the start, to practical realization of those prospective cash flows. Then capital growth arrives when the market first evaluates prospective cash flows, and is realized eventually in physical outcomes insofar as the market has predicted correctly. That is to say that capital growth, in the mind of the market, precedes and pays for any needed investment in new plant and equipment.[3] Early investors borrow from the future, in effect, and pay the interest from increments in present value as positive cash flows approach with time.

Meanwhile the innovator acquires materials and plant capacity and labor skills at market prices determined by their uses in current technology, but applies them more productively until competition catches up. It is that temporary market

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[1]As reported in national accounts; saving and investment differ only by statistical discrepancy. We view them as different, for reasons shown in Appendix A, but will equate them in this paper by using the word "saving" in the Keynesian sense of invested saving.

[2]We argue, however, that productivity gain will not give equal capital growth unless gross saving is enough to offset depreciation. We interpret any shortfall in this coverage as a negative component in net saving, and expect capital growth to equal productivity gain less that deficit (see Section 3).

[3]Present value is time-discounted, and affords innovators only enough, at first, for first steps in implementation. As implementation proceeds, present value rises with time and affords more.

advantage to the innovator, which we call the innovator's reserve, that explains capital growth without net saving in a practical and mechanical sense, while the present value principle gives the source of investment. This idea will be called free growth theory for easy reference.

It predicts only at the largest scales, and only for the private sector. Individuals and groups and even small economies can grow in capital through saving if others dissave equally. That possibility is foreclosed only at the scale of all capital and all economies together. The public sector, meanwhile, responds to political rather than market choices, and grows or shrinks accordingly.

If free growth theory is right, tax policy and other policy to encourage saving over consumption should be reviewed. These policies include the higher tax on ordinary income than on capital gains, and the double tax on corporate dividends.

Inferences for economic teaching include the obvious ones for growth theory and for net saving in general. They include others as well. One of the central doctrines of the marginalist revolution has held that market realization converges to producer cost, when that cost includes imputed interest on assets owned. Net saving gives producer cost, and falls short of market realization in the presence of technological growth from new ideas. Meanwhile the doctrine that net income equals consumption plus net saving is put into question by evidence offered here suggesting that net saving increases the physical quantity of capital, but not the aggregate value.

## 2. Net saving, capital growth and capital acceleration

Thrift theory will mean the combined ideas that net saving $S_{net}$ is realized in equal growth of capital $K$, if $S_{net}$ holds within the warranted rate, and that there is no other source of capital growth except market noise regressing to zero. "Thrift assumptions" will mean those two plus the assumptions that current $S_{net}$ holds within the warranted rate. The acronym uta will mean "under thrift assumptions." Then

$$\Delta K = S_{net} \,, \quad \text{uta.} \tag{1}$$

Define "thrift" $s^*$ by $s^* = \frac{S_{net}}{K}$, and divide Eq. (1) by capital to predict capital growth rate $g(K)$ as

$$g(K) = s^* \,, \quad \text{uta,} \quad \text{where } s^* = \frac{S_{net}}{K}. \quad \text{Then} \tag{2}$$

$$\Delta g(K) = \Delta s^* \,, \quad \text{uta.} \tag{3}$$

$\Delta g(K)$ and $\Delta s^*$ respectively will be called "capital acceleration" and "thrift change," either of which may be positive or negative or zero. Division of Eq. (3) by capital acceleration, with rearrangement, gives

$$\frac{\Delta s^*}{\Delta g(K)} = \frac{\Delta g(K)}{\Delta g(K)} = 1 \,, \quad \text{uta.} \tag{4}$$

Algorithms for correlation and covariance and regression are structured on the reasoning that causal relationships among variables tend to be revealed in concurrent changes. While Eqs. (1) and (2) enable measurement of the coexistence of net saving and capital growth, Eqs. (3) and (4) allow a better measure of the degree to which that coexistence is explained by causality. Eq. (4) was derived from Eq. (3) so that success of predictions from thrift assumptions can be measured against a standard of unity. For notational convenience, then, we define the "thrift index" $\theta$ by $\theta = \frac{\Delta s^*}{\Delta g(K)}$, and restate Eq. (4) as

$$\theta = 1 \,, \quad \text{uta,} \quad \text{where } \theta = \frac{\Delta s^*}{\Delta g(K)} \,. \tag{5}$$

Test results for the predictions $\frac{s^*}{g(K)} = \theta = 1$, under thrift assumptions, appear in Fig. 1 and in Tables 1, 2 and 3. Screens to which they refer are explained in Section 16 below. Results were found as GDP-weighted averages over all countries and years, from data for net saving and market-value capital taken from national accounts. Figures 1 and 2 show $\frac{s^*}{g(K)}$ and $\frac{\Delta s^*}{\Delta g(K)}$ as 0.481 and 0.064 respectively. Table 1 shows regression of $s^*$ on $g(K)$ as 0.0771, and regression of $\Delta s^*$ on $\Delta g(K)$ as 0.0559.

**Table 1**
Regression of $s^*$ on $g(K)$ and $\Delta s^*$ on $\Delta g(K)$, GDP-weighted. Screen = 0.01. $H_0$ per thrift theory: $\frac{s^*}{g(K)} = \frac{\Delta s^*}{\Delta g(K)} = 1$.

| | | | |
|---|---|---|---|
| Regression of $s^*$ on $g(K)$ | 0.0771*** | Regression of $\Delta s^*$ on $\Delta g(K)$ | 0.0559*** |
| | (0.0144) | | (0.0057) |
| Observations | 1,826 | Observations | 1,574 |
| $R^2$ | 0.84511 | $R^2$ | 0.37071 |
| Within $R^2$ | 0.10669 | Within $R^2$ | 0.27233 |
| Year fixed effects | ✓ | Year fixed effects | ✓ |
| Country fixed effects | ✓ | Country fixed effects | ✓ |

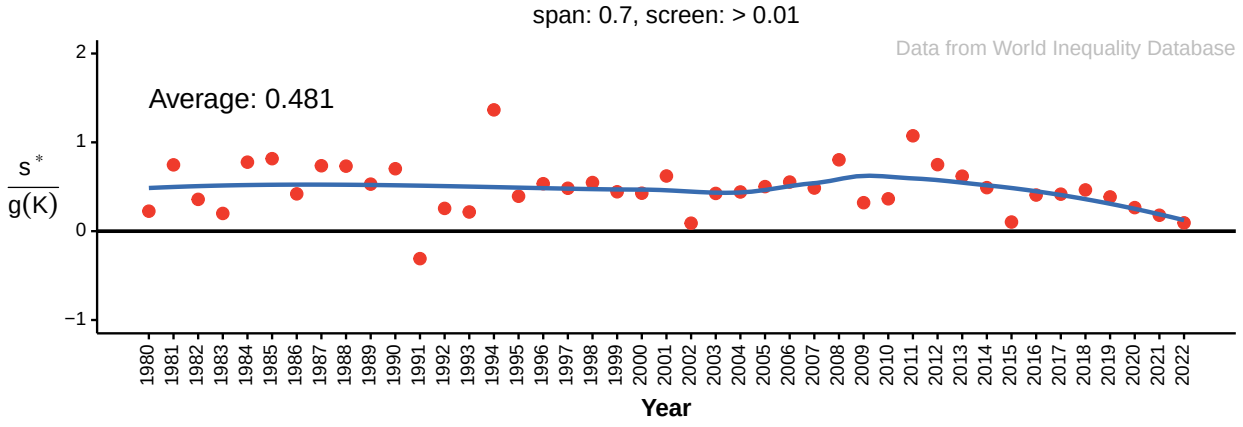



**Fig. 1:** Average $\frac{s^*}{g(K)}$ with LOESS smoothing over all countries, GDP-weighted, 1980-2022

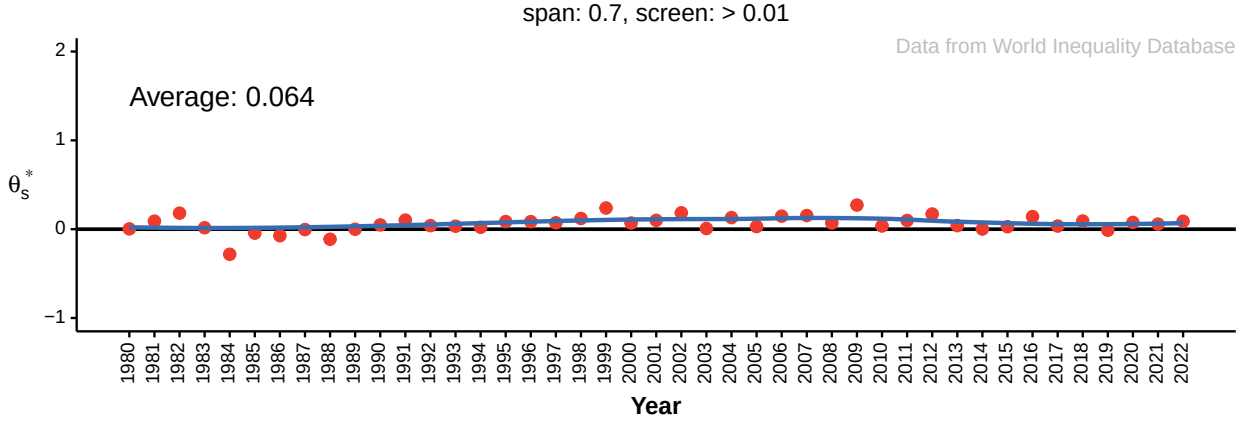



**Fig. 2:** Average $\theta_s^*$ with LOESS smoothing over all countries, GDP-weighted, 1980-2022

**Table 2**
Average $\frac{s^*}{g(K)}$ in 92 countries (screen = 0.01). Number of years clearing screen shown in ()

| | | | | | |
|---|---|---|---|---|---|
| Armenia | 1996 - 2020 (23) | -0.63 | Italy | 1980 - 2022 (36) | 0.13 |
| Aruba | 1996 - 2001 (6) | 1.25 | Japan | 1980 - 2021 (29) | 0.19 |
| Australia | 1980 - 2018 (33) | 0.24 | Kazakhstan | 1997 - 2022 (25) | 0.60 |
| Austria | 1996 - 2021 (23) | 0.83 | Kuwait | 2003 - 2017 (14) | 1.11 |
| Azerbaijan | 1996 - 2020 (23) | 1.57 | Kyrgyzstan | 1996 - 2021 (24) | 0.46 |
| Bahrain | 2009 - 2013 (5) | -2.43 | Latvia | 1996 - 2021 (24) | -0.23 |
| Belgium | 1996 - 2021 (21) | 0.61 | Lithuania | 1996 - 2021 (24) | 0.13 |
| Bolivia | 1998 - 2015 (17) | 0.40 | Luxembourg | 1996 - 2021 (22) | 0.03 |
| Botswana | 1996 - 1999 (4) | 3.54 | Malaysia | 2007 - 2015 (9) | 2.52 |
| Brazil | 1996 - 2019 (21) | 0.25 | Malta | 1996 - 2021 (23) | 0.16 |
| British Virgin Islands | 1996 - 1999 (4) | 2.45 | Mauritius | 2014 - 2018 (5) | 0.04 |
| Bulgaria | 1996 - 2016 (16) | 0.14 | Mexico | 1996 - 2021 (22) | 0.15 |
| Burkina Faso | 2000 - 2019 (20) | 0.23 | Moldova | 1996 - 2019 (22) | -0.91 |
| Cameroon | 1996 - 2019 (24) | 1.10 | Mongolia | 2006 - 2020 (15) | 0.10 |
| Canada | 1980 - 2022 (38) | 0.37 | Morocco | 1999 - 2021 (23) | 1.13 |
| Cape Verde | 2008 - 2017 (9) | 0.53 | Netherlands | 1996 - 2021 (23) | 0.51 |
| Chile | 1997 - 2021 (19) | 0.84 | New Zealand | 1996 - 2019 (21) | 0.78 |
| China | 1992 - 2020 (29) | 0.81 | Nicaragua | 2007 - 2018 (12) | 0.08 |
| Colombia | 1996 - 2022 (27) | 0.60 | Niger | 1996 - 2019 (24) | 0.90 |
| Costa Rica | 2013 - 2020 (6) | 0.21 | Norway | 1981 - 2021 (38) | 0.95 |
| Croatia | 1996 - 2021 (24) | 0.43 | Peru | 2008 - 2021 (14) | 0.76 |
| Curaçao | 2001 - 2016 (14) | 1.26 | Philippines | 1996 - 2022 (27) | 1.76 |
| Cyprus | 1996 - 2021 (25) | 0.07 | Poland | 1996 - 2021 (25) | 0.71 |
| Czechia | 1994 - 2020 (20) | 0.17 | Portugal | 1996 - 2022 (24) | 0.00 |
| Côte d'Ivoire | 1996 - 2000 (5) | 0.29 | Qatar | 2002 - 2017 (14) | 1.89 |
| Denmark | 1996 - 2022 (26) | 0.22 | Romania | 1996 - 2020 (22) | 0.03 |
| Dominican Republic | 1996 - 2016 (10) | 1.52 | Russia | 1996 - 2017 (13) | -0.19 |
| Ecuador | 2008 - 2020 (13) | 0.53 | Saudi Arabia | 2003 - 2009 (7) | 2.65 |
| Egypt | 1997 - 2015 (19) | 0.92 | Senegal | 2015 - 2021 (7) | 0.24 |
| El Salvador | 2015 - 2019 (5) | -0.62 | Serbia | 1998 - 2021 (23) | 0.61 |
| Estonia | 1996 - 2021 (24) | 0.45 | Slovakia | 1996 - 2022 (24) | 0.15 |
| Finland | 1996 - 2021 (23) | 0.17 | Slovenia | 1996 - 2021 (20) | 0.28 |
| France | 1980 - 2021 (35) | 0.26 | South Africa | 1996 - 2022 (24) | 0.28 |
| Germany | 1980 - 2022 (38) | 0.70 | South Korea | 1996 - 2020 (24) | 0.63 |
| Greece | 1996 - 2021 (26) | 0.17 | Spain | 1995 - 2021 (25) | 0.32 |
| Guatemala | 2002 - 2021 (20) | -0.72 | Sweden | 1980 - 2021 (37) | 0.41 |
| Guinea | 2004 - 2010 (5) | 0.72 | Switzerland | 1993 - 2021 (25) | 0.69 |
| Honduras | 2001 - 2015 (14) | -0.57 | Tunisia | 1996 - 2011 (16) | 0.29 |
| Hong Kong | 1997 - 2011 (15) | 0.90 | Turkey | 2010 - 2017 (8) | 1.32 |
| Hungary | 1996 - 2021 (21) | 0.14 | USA | 1980 - 2021 (38) | 0.31 |
| Iceland | 2001 - 2014 (14) | 0.22 | Ukraine | 1996 - 2019 (21) | 0.17 |
| India | 1999 - 2019 (21) | 0.57 | United Kingdom | 1981 - 2021 (36) | 0.16 |
| Indonesia | 2017 - 2019 (3) | 0.94 | Uruguay | 2016 - 2017 (2) | 1.16 |
| Iran | 1997 - 2017 (20) | 4.96 | Uzbekistan | 2011 - 2021 (10) | 0.21 |
| Ireland | 1996 - 2021 (22) | 0.26 | Vanuatu | 2002 - 2007 (6) | 0.29 |
| Israel | 1996 - 2019 (24) | 0.46 | Venezuela | 1998 - 2019 (20) | 0.55 |

*Note:* $s^*$ is defined as $\frac{S_{net}}{K}$. Thrift theory predicts $\frac{s^*}{g(K)} = 1$. Free growth theory makes no prediction for these data.

**Table 3**
Average $\theta$ in 92 countries (screen = 0.01). Number of years clearing screen shown in ()

| | | | | | |
|---|---|---|---|---|---|
| Armenia | 1997 - 2020 (20) | 0.12 | Italy | 1972 - 2021 (29) | 0.05 |
| Aruba | 1997 - 2001 (5) | 1.22 | Japan | 1972 - 2021 (38) | 0.05 |
| Australia | 1974 - 2019 (35) | 0.03 | Kazakhstan | 1997 - 2022 (23) | 0.08 |
| Austria | 1997 - 2021 (16) | 0.10 | Kuwait | 2005 - 2017 (11) | 0.04 |
| Azerbaijan | 1997 - 2020 (23) | 0.75 | Kyrgyzstan | 1998 - 2021 (22) | 0.23 |
| Bahrain | 2010 - 2013 (4) | 0.41 | Latvia | 1998 - 2021 (20) | 0.01 |
| Belgium | 1997 - 2021 (19) | 0.05 | Lithuania | 1997 - 2021 (18) | 0.10 |
| Bolivia | 1999 - 2014 (12) | 0.04 | Luxembourg | 1997 - 2021 (25) | 0.00 |
| Botswana | 1997 - 2000 (4) | 0.24 | Malaysia | 2008 - 2015 (7) | 0.01 |
| Brazil | 1998 - 2020 (22) | 0.10 | Malta | 1997 - 2021 (23) | 0.08 |
| British Virgin Islands | 1997 - 1998 (2) | 2.24 | Mauritius | 2015 - 2018 (4) | 0.02 |
| Bulgaria | 1997 - 2017 (15) | -0.03 | Mexico | 1997 - 2021 (22) | 0.06 |
| Burkina Faso | 2001 - 2019 (16) | 0.04 | Moldova | 1997 - 2019 (21) | 0.11 |
| Cameroon | 1997 - 2019 (19) | 0.18 | Mongolia | 2008 - 2021 (11) | 0.10 |
| Canada | 1974 - 2022 (41) | 0.09 | Morocco | 2000 - 2021 (17) | 0.19 |
| Cape Verde | 2009 - 2016 (8) | 0.27 | Netherlands | 1997 - 2021 (20) | 0.03 |
| Chile | 1998 - 2021 (22) | 0.06 | New Zealand | 1997 - 2019 (18) | 0.14 |
| China | 1981 - 2020 (33) | 0.07 | Nicaragua | 2008 - 2019 (12) | 0.06 |
| Colombia | 1997 - 2022 (18) | -0.03 | Niger | 1997 - 2019 (19) | 0.22 |
| Costa Rica | 2014 - 2020 (6) | 0.04 | Norway | 1982 - 2021 (34) | 0.12 |
| Croatia | 1997 - 2021 (22) | 0.19 | Peru | 2009 - 2021 (11) | 0.04 |
| Curaçao | 2002 - 2016 (14) | 0.41 | Philippines | 1997 - 2022 (20) | 0.37 |
| Cyprus | 1997 - 2021 (23) | -0.06 | Poland | 1997 - 2021 (15) | 0.20 |
| Czechia | 1995 - 2021 (19) | -0.03 | Portugal | 1997 - 2022 (17) | 0.02 |
| Côte d'Ivoire | 1997 - 2000 (4) | -0.03 | Qatar | 2004 - 2018 (12) | 0.29 |
| Denmark | 1997 - 2022 (22) | 0.05 | Romania | 1997 - 2020 (14) | 0.02 |
| Dominican Republic | 2007 - 2015 (9) | 0.15 | Russia | 1997 - 2018 (14) | 0.09 |
| Ecuador | 2009 - 2020 (12) | 0.17 | Saudi Arabia | 2005 - 2009 (3) | 2.23 |
| Egypt | 1998 - 2015 (16) | 0.13 | Senegal | 2018 - 2021 (3) | 0.19 |
| El Salvador | 2016 - 2019 (4) | 0.98 | Serbia | 1999 - 2021 (22) | 0.22 |
| Estonia | 1997 - 2021 (19) | 0.06 | Slovakia | 1997 - 2021 (18) | 0.09 |
| Finland | 1997 - 2022 (21) | -0.02 | Slovenia | 1997 - 2021 (20) | 0.10 |
| France | 1972 - 2021 (33) | 0.09 | South Africa | 1997 - 2022 (21) | 0.10 |
| Germany | 1972 - 2022 (31) | 0.05 | South Korea | 1997 - 2020 (15) | 0.04 |
| Greece | 1997 - 2021 (24) | 0.24 | Spain | 1972 - 2021 (42) | 0.06 |
| Guatemala | 2003 - 2021 (11) | 0.15 | Sweden | 1974 - 2022 (41) | 0.02 |
| Guinea | 2005 - 2010 (6) | 0.08 | Switzerland | 1993 - 2020 (24) | 0.08 |
| Honduras | 2003 - 2015 (10) | 0.40 | Tunisia | 1997 - 2011 (12) | 0.32 |
| Hong Kong | 1997 - 2011 (13) | 0.04 | Turkey | 2011 - 2015 (5) | -0.16 |
| Hungary | 1997 - 2021 (21) | 0.07 | USA | 1972 - 2021 (44) | 0.02 |
| Iceland | 2002 - 2014 (12) | 0.10 | Ukraine | 1997 - 2020 (23) | 0.05 |
| India | 1997 - 2019 (16) | 0.14 | United Kingdom | 1972 - 2021 (42) | 0.02 |
| Indonesia | 2019 - 2019 (1) | -0.41 | Uruguay | 2017 - 2017 (1) | 0.56 |
| Iran | 1998 - 2018 (19) | 1.09 | Uzbekistan | 2012 - 2021 (9) | 0.05 |
| Ireland | 1997 - 2021 (23) | 0.02 | Vanuatu | 2003 - 2007 (3) | 0.15 |
| Israel | 1997 - 2017 (18) | 0.07 | Venezuela | 1999 - 2019 (20) | 0.31 |

*Note:* $\theta$ is defined as $\frac{\Delta s^*}{\Delta g(K)}$. Thrift theory predicts $\theta = 1$. See Section 3 for predictions of free growth theory.

## 3. Interpretation of test results

The finding $\frac{s^*}{g(K)} = 0.481$ reveals average observed proportions between $s^*$ and $g(K)$, we repeat, and does not of itself show causal relatedness between those two variables. Causal relatedness is shown rather by the degree of constancy in proportions, and thus on regression of $s^*$ on $g(K)$ found at 0.0771.

$\theta$ or $\frac{\Delta s^*}{\Delta g(K)}$, measured at 0.064, gives a separate measure of causal relatedness between $s^*$ and $g(K)$ as explained. Regression of $\Delta s^*$ on $\Delta g(K)$, found at 0.0559, gives a measure of causal relatedness between $\Delta s^*$ and $\Delta g(K)$. Might these small but positive findings allow the possibility that some capital growth is explained by net saving?

Free growth theory predicts that the present value of new ideas can pay for equal capital growth, but only after gross saving has made up for all depreciation. In years of recession, households may defer part or all of depreciation saving to support consumption. In these years, growth in the present value of better ideas will not realize equal net growth of capital, as some of it must offset any shortfall in depreciation saving first. Allowing for these years, we can say more compactly that free growth theory predicts that a gain in the present value of new ideas accounts for all capital creation, and thus for the sum of capital growth and restoration of shortfalls in depreciation saving. Thus free growth theory does not predict a zero value for $\theta$, but rather predicts $\theta$ to equal the share of capital creation applied to restoring those shortfalls. With this understanding, free growth theory could fit the appearance in Fig. 2 that $\theta$ found higher values during the period of the Iraq-Kuwait, dot com and subprime crises. We will suppose that 0.064 gives a plausible estimate of the shortfall in depreciation saving, until more is known, and will proceed on the simplifying assumption that free growth theory predicts correctly.

## 4. Inferences for the creative destruction idea

Creative destruction (Schumpeter (1912)[4]) is obsolescence of existing capital through innovation of new and preferred technology, as with horse-drawn carriages and automobiles. Granting the obsolescence of old ways by new, it does not follow that innovation reduces values of assets already present. Data cited in this paper appear to show that capital growth is maximized when investment is enough to offset depreciation. A natural source of that investment is depreciation of assets the new technology will replace, and a natural market for the new technology may be owners of the assets to be replaced. Carriage owners were better able to afford cars when the carriages in which they had invested before lived out their economic lives as originally expected, so that the depreciation component in the cash flows originally expected can pay for cars in replacement just as it might have paid for new carriages.

This analysis suggests that market forces favor a kind through orderly obsolescence through replacement without premature displacement as the new technology awaits its turn.

## 5. Solow's puzzle

Some growth is capital widening, where structures and implements increase in number but do not change in design. Capital widening, however, is practical only so far before glut and diminishing returns set in. Further growth must come from capital deepening, meaning improvements in the design of capital. Solow (1956) noted a kind of middle ground between capital widening and capital deepening in the disembodied growth mentioned earlier; ships carrying coals to Newcastle can raise prospective cash flows, and hence present value, by reversing the business plan. But Solow, who came to conclusions similar to ours from different evidence, puzzled as to how capital growth without net saving could be possible for capital deepening through "embodied" growth, where products of new design are made from plant of new design.

Solow received the Nobel Prize in 1987. His acceptance speech mentions the embodiment problem:[5]

"... much technological progress, maybe most of it, could find its way into actual production only with the use of new and different capital equipment. Therefore the effectiveness of innovation in increasing output would be paced by the rate of gross investment. A policy to increase investment would therefore lead to ... a faster transfer of new technology into actual production ... By 1958 I was able to produce a model that allowed for the embodiment effect. If common sense was right, the embodiment model should have fit the facts better than the earlier one. But it did not... I do not know if that finding should be described as a paradox, but it was at least a puzzle."

[4]Published by Schumpeter in 1912, and Translated by Opie in 1934.

[5]Solow (1988). The terms capital deepening, capital widening, embodied growth and disembodied growth are all Solow's.

Solow was right in that much innovation requires months or years of gross investment to pay for negative cash flows in building new plant before the new products roll out from the factory doors. What he left unsaid is that this investment would not have been forthcoming unless positive cash flows at the end of the tunnel were predicted to be sufficient to create positive present value from the start. Data in our displays suggest that this initial creation of value, plus depreciation investment, are enough.

## 6. Mechanics of free growth

The present value argument just given offers an accounting explanation of embodied growth in terms of sources of funds. The step-by-step mechanical explanation, we suggest, in terms of things rather than funds, is that embodied growth is disembodied growth on a finer scale. At each step toward realization of the new plant and products, raw materials and products and labor skills and plant capacity currently available on the market are adapted to new uses. The innovator pays for these inputs at a market price determined by their value in established productive uses, but applies them innovatively to realize higher prospective cash flows, and hence higher present values, to the innovator (Marshall (1890), Schumpeter and Opie (1934)). This difference in present value realized less price paid will here be called the "innovator's reserve", meaning reserve price for inputs of capital and labor.[6] The innovator's reserve quantifies the part of free growth explained by productivity gain as distinct from random market noise. As such, it is the quantity added to depreciation saving to enable embodied growth, so that net saving is never needed.

Our findings support those of Piketty and Zucman (2014) and Kurz (2023) as to the market power of innovators to explain capital growth beyond net saving. Those studies assumed that net saving is recovered in capital growth, so that this market power, which we call the innovator's reserve, need explain only the part of capital growth remaining. Again, we go farther by questioning the assumption that net saving contributes even a part of capital growth.

## 7. Tech Externalities

The innovator's reserve, quantified by the present value of the innovator's idea, gives the amount of free growth to the innovator. Others may benefit indirectly. Old factories and warehouses and residences gain in utility as they house higher-tech equipment and worker skills. Old highways do the same as they carry higher-tech freight and drivers and passengers to higher-tech destinations in higher-tech vehicles.[7]

Most externalities,[8] positive and negative, are part of random market noise expected to offset to zero over time and scale. The examples above, which we call tech externalities, can impose a prevailingly positive vector resisting the downward pull of depreciation. Free growth, in the full sense, means effects of the innovator's reserve on capital of the innovator, plus effects of tech externalities on capital in general.

## 8. Optimum investment policy

Data and arguments suggest that the optimum amount of saving, at the global scale, is replacement saving to offset depreciation, and nothing more. That would not mean book depreciation, as this study has stressed differences between book and market values. Up to a point, it should be possible to analyze the composition of market capital, and to model depreciation of the whole. A better plan, as Solow (1956) wrote in response to Harrod's knife edge argument (Harrod, 1939), is to trust the market to maximize rate of return, and to sense the point where glut begins and returns fall.[9] Markets do so imperfectly when tax and other public policy reward saving over distributions and consumption. Findings in this paper suggest review of such policies. These include the double tax on dividends, and the greater tax rate on ordinary income than on capital gains. Effects of removing the double tax, and removing the difference between tax rates on ordinary income and on capital gains, could be revenue-neutral and non-partisan if the corporate tax were raised to match, if the tax rates on ordinary income and on capital gains met somewhere between, and if thoughtful grandfathering eased the transition.

[6] i.e., capital and labor inputs are worth more to the innovator in that the innovator applies them in ways to realize greater returns. The present value of additional cash flow enabled by this advantage in return quantifies the innovator's reserve, and equivalently the non-random component of free growth.

[7] Rae (1834) makes essentially the same argument in terms of the technology of his time.

[8] Externalities are defined as effects of events on nonparticipants.

[9] Harrod had argued that saving must hit the warranted rate exactly or risk positive feedback through the operation of the output/capital ratio (accelerator).

## 9. Optimum government policy

Households dissave in consumption, while firms dissave through dividends and other distributions. Public policy, in many nations, has inhibited dissaving in the belief that such policies favor capital growth. Rather they inhibit consumption, add nothing to capital growth, and thus inhibit net output as a function of both together (see Appendix B).

The effect of repealing such policies would be to address capital glut not by investing less, but by disinvesting more.[10] This strategy spares government the burden of guiding investment, at least until the effects of the repeal are known.

## 10. Inferences for national accounts and macroeconomics

We have attempted to show that net saving does not tend to equate to capital growth, and indeed contributes no part of capital growth. If our argument proves correct, then national accounts, and the teachings of macroeconomics (macro) which they express, are constructed on a fundamental misperception. We would therefore welcome any critiques of our findings as a matter of academic and public interest.

## 11. Inferences for microeconomics

Microeconomics ("micro") studies the mutual effects of supply, demand and price. It teaches that producer cost, including interest as the cost of time, tends to equilibrate to market value realized. We have proposed an exception in the innovator's reserve; the innovator pays the going rates for skills and materials needed, but applies them to make products of greater market value.

## 12. Inferences for inflation

If free growth theory is correct, then net investment adds a superfluous cost to capital growth, while decreasing market value per unit through capital glut. Both effects would tend to contribute directly to capital inflation, and indirectly to inflation of consumer goods by adding depreciation costs. This consideration adds to the case for urgent attention to our argument, and for more study to confirm or refute it.

## 13. Net output

In Keynesian tradition, net output equals consumption plus net investment. We have disputed this reasoning in our argument that net investment adds nothing to capital growth, inhibits consumption, and thus tends to reduce the sum of both. We also critique it on another ground. Assume, for the sake of argument, that we are wrong, and that net investment realizes equal capital growth. Net output means value added. Appendix B will show an argument that to interpret value added as the sum of consumption and capital growth, and neither less nor more, would miss two components, one negative (human depreciation) and one positive (self-invested work), in value added to human capital. Both of these flows are unmeasurable by known means. Thus we have not sought to measure net output or its growth, and have treated capital growth alone.

## 14. Data sources

All our data are drawn from the free online website World Inequality Database (WID). This source collates data from national accounts and tax data of 105 countries in constant currency units, and adjusts them where needed to conform to current standards of the System of National Accounts (SNA) published by the United Nations. We show results for the 92 of those countries which report both factors, net saving and market-value capital, needed for deriving the thrift indexes. WID's source for these data is national accounts.

Net saving $S_{net}$ and market-value $K$ are taken from net national saving (msavin) and market-value Capital Wealth (mnweal) respectively. GDP, which we use only for weighting purposes in Figs. 1 and 2, is reproduced from GDP (mgdpro).

[10] Higher dividends and distributions buy firms, that is, would enable higher consumption by households.

## 15. Accessing our results and methods

Tables and other displays of our findings for each country, and showing our methods of calculation, can be accessed at the web appendix (https://web-appendix.shinyapps.io/Sources_of_Capital_Growth).

## 16. Displays

The web appendix includes displays of $\theta$ for each year in each country over the report period. These tend to show upward and downward spikes in values of $\theta$ in some years. Those spikes tend to be associated with small absolute values of denominators, in these cases $\Delta g(K)$, in those countries and years. Small denominators magnify errors in measurements of numerators. Worse, when denominators are small, small mismeasurements in them might reverse the term in sign.

To maximize reliability of test results, we apply a range of screens to omit years where absolute denominators fall below a given threshold. Some displays in the web appendix show data for all years, regardless of denominator size. Others screen out all years where absolute denominators are less then 0.01, then 0.025, then continuing upward in increments of 0.025 to a maximum screen of 0.15. $\theta$ is plotted for each country unscreened and at each of the seven successive levels of screening. All displays in the main body of this paper apply a screen of 0.01. The denominator whose absolute value is screened is capital acceleration $\Delta g(K)$ or capital growth rate $g(K)$ in all displays.

Screening out years where absolute $\Delta g(K)$ or $g(K)$ is small would cost little in informative value even if measurements were exact. In those years, there is little capital acceleration or capital growth, positive or negative, for either thrift theory or free growth theory to explain. Market noise alone might account for $\Delta g(K)$ or $g(K)$ in such years. Screening reduces the number of observations, but increases the reliability and informative value of each.

## 17. Disclaimers

We accept that capital growth is impossible without capital replacement first, to make up losses to depreciation, and that the expected source of capital replacement is replacement investment. Thus we accept the necessity and efficacy of replacement investment. We dispute only the efficacy of net investment in realizing capital growth after replacement investment has made up for depreciation.

Evidence supports the proposal of Modigliani (1954) that the young save to support retirement, and that retirees dissave to support consumption.[11] Such age-based differences are expected to offset in the stationary state, where tradition and free growth theory agree that overall net saving is zero. We accept that individual net saving can realize equal individual capital growth, so long as collective saving does not exceed collective dissaving.

## 18. Discussion and conclusions

Capital glut is the condition warned against by West, Ricardo, Malthus and Harrod. It is loosely defined as oversupply of capital at the current state of technology. We take it to mean or include a state where more investment does not realize more capital value. Findings shown in our displays suggest that net saving raises the physical quantity of capital, say in number of shops, manufacturing plants or finished goods of similar design, without raising aggregate value of capital, and so contributes to capital glut by that understanding.

These findings challenge the teachings that capital growth is effected by net saving enabled by consumption restraint, and that producer cost, including imputed interest as the opportunity cost of capital, converges to market realization. Evidence suggests that all capital growth is free, and consequently that market realization, in the presence of innovation, exceeds producer cost by the entirety of capital growth. Meanwhile the same evidence, which indicates that net saving adds no capital value, suggests a review of the teachings that consumption plus net saving gives net income, and that consumption plus net investment gives net output. Appendix B will also question the latter teachings.

Embodied growth is disembodied growth on a finer scale. It redeploys or repurposes existing labor skills, raw materials, and plant capacity, as well as existing finished goods, to achieve higher returns than available from the customary uses which determine their prices. The present value of yields from this advantage in return, or equivalently the innovator's reserve, defines the non-random component in free growth.

[11] Friedman (1957) follows a different argument to a similar conclusion.

Solow's puzzle runs deep. How can thoughts be things? How can assessments of present value pay, at least in part, for their own realization? Science and philosophy push our minds into areas where they are not at home. Neither relativity nor quantum mechanics, which explore the scale of great speeds and minute distances, sink in easily. Economics, too, invites surprises as it looks farther from the scale of the household and community and firm to which we are accustomed. Free growth theory predicts only at the universal scale, and accepts that thrift explains a greater part of growth at progressively smaller scales. Our findings, with that allowance, are awkward for much of economics as generally taught. We encourage others to check our data and methods and inferences, and improve on them wherever they can.

## Appendix A. Equivalence of saving and investment

Keynes (1936) argued correctly that saving, if not invested in currently produced producer goods and services, adds nothing to capital, and thus nothing to output. He inferred that uninvested saving does not achieve actual saving, as it accomplishes no growth, and described it as "intended saving." True saving, by that reasoning, equalled investment.

We suggest that investors seek to maximize return, within risk tolerance, and will sometimes hold saving in cash or equivalents at zero return in recessions and depressions when positive returns cannot be found, and so when investing would tend to reduce capital and output rather than increase them. Thus we do not share the Keynesian view that saving need be invested in order to maximize its value. Saving will tend to equal investment in normal times, when prospective returns bring animal spirits, and could include saving in cash when not. It is only for these profitable times that we accept the equality of saving and investment, and only as a simplifying assumption.[12]

## Appendix B. Net output with human capital

Human capital is impractical to measure, as it leaves little market record other than for its rental income in pay and investment cost in nurture and schooling. Thus national accounts leave it implicit, and allow us to infer what we can from data for pay and nurture and schooling. Those accounts are founded on the principle, sound when terms are appropriately redefined, that net output, or value added, is expressed in the sum of capital growth and net outflow from the value-added chain. In national accounts, then, where physical capital is the whole of capital while net outflow of the chain is the whole of consumption, the reasoning is

$$Y = \Delta K + C\,, \quad \text{neglecting human capital.} \tag{B.1}$$

It is possible in principle to model a value-added chain which includes human capital, and to compare findings with those shown in Eq. (B.1). Let human capital $H$, in that new model, stand as the last link in the value-added chain. Adapting the classic illustration of the value added principle, say that farms produce wheat, mills convert the wheat to flour, bakeries convert the flour into bread, and humans convert some of the bread, called invested consumption, into human capital. The net outflow from this extended value-added chain is not all of consumption, but only pure consumption, meaning the part remaining after the part invested in human capital (invested consumption) is subtracted.[13] By this reasoning, the principle that net output is expressed in capital growth plus net outflow gives

$$Y = \Delta K + \Delta H + C_p\,, \quad \text{allowing human capital,} \tag{B.2}$$

where $C_p$ gives pure consumption.

Yoram Ben-Porath (1967) reasoned that growth in human capital equals invested consumption plus self-invested work less human depreciation[14]. Let $C_s$, $W_s$ and $D(H)$ show these flows respectively. Thus the combined arguments of Schultz and Ben-Porath arrive at

$$C = C_s + C_p \quad \text{and} \quad \Delta H = C_s + W_s - D(H)\,, \quad \text{allowing human capital.} \tag{B.3}$$

Substitution of these equations into Eq. (B.2) finds

$$Y = \Delta K + C_s + W_s - D(H) + C_p \quad \text{and consequently}$$

[12] The weakness in the Keynes argument, if we are right, was his inference that saving does not accomplish its purpose unless invested to realize capital growth. Its purpose, we argue, is to maximize capital growth according to investment opportunities, and to maximize it by withholding investment when all investments promise loss.

[13] The concepts of invested consumption, pure consumption, self-invested work and human depreciation were introduced in Schultz (1961).

[14] Equation 4 in Ben-Porath's paper, summarizing his first three equations. His terms and notation differ from ours.

$$Y = \Delta K + C + W_s - D(H)\,, \quad \text{allowing human capital,} \tag{B.4}$$

if Schultz and Ben-Porath and the reasoning here are right.

## Appendix C. Mill's statement of the free growth idea

Mill (1848), book 1, chapter 5, section 4, includes:

> If it were said, for instance, that the only way to accelerate the increase of capital is by increase of saving, the idea would probably be suggested of greater abstinence, and increased privation. But it is obvious that whatever increases the productive power of labour creates an additional fund to make savings from, and enables capital to be enlarged not only without additional privation, but concurrently with an increase of personal consumption.

This passage may be the first clear statement of what we call free growth theory. Mill's use of the words "accelerate" and "concurrently" suggest that his path of reasoning was something like ours, and that he meant what we call free growth rather than the alternating phases of higher and lower saving rates pictured in thrift theory.

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